2024-12-03-Duality

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Slide 1

Content

Duality
(P) max $c^T x$
s.t. $Ax \leq b$
$x \geq 0$
$\xrightarrow{y}$ $y^T Ax \leq y^T b$
$c^T x \leq y^T Ax \leq y^T b$
$c^T \leq y^T A$
(D) min $b^T y$
s.t. $A^T y \geq c$
$y \geq 0$

Description

Slide 1 of 38 introduces the concept of duality in linear programming. It presents a primal linear programming problem (P) and its corresponding dual problem (D). The primal problem (P) is a maximization problem with objective function $c^T x$ subject to constraints $Ax \leq b$ and $x \geq 0$. The dual problem (D) is a minimization problem with objective function $b^T y$ subject to constraints $A^T y \geq c$ and $y \geq 0$. The arrow indicates the derivation of the dual problem from the primal problem. The intermediate steps show how the inequality $c^T x \leq y^T Ax \leq y^T b$ is obtained by multiplying the primal constraint $Ax \leq b$ by y^T (where $y \geq 0$), and then using this to derive the dual constraint $c^T \leq y^T A$. The red boxes highlight the primal and dual problem formulations. The context is the introduction to duality theory in linear programming, which is a fundamental concept for solving and analyzing linear programming problems.

Figures

(a)

Slide 2

Content

Complementary Slackness (at optimal)
c$^T x = y^T Ax = y^T b$
$c^T x = y^T Ax$
$0 = (y^T A - c^T)x$
$\underbrace{z}_{0=z^T x}$
$x_j z_j = 0$
$y^T Ax = y^T b$
$y^T (Ax - b) = 0$
$\underbrace{w}_{y^T w=0}$
$y_i w_i = 0$

Description

Slide 2 of 38 introduces the concept of complementary slackness in linear programming. It states that at optimality, the primal objective function value ($c^T x$) equals the dual objective function value ($y^T b$), which in turn equals $y^T Ax$. This equality is then used to derive two key conditions. The first condition, $0 = (y^T A - c^T)x$, is obtained by subtracting $c^T x$ from $y^T Ax$. The vector $z = y^T A - c^T$ represents the reduced costs, and the equation $0 = z^T x$ implies that for each variable x_j, either $x_j = 0$ or the corresponding reduced cost $z_j = 0$ (or both). This is expressed concisely as $x_j z_j = 0$. Similarly, the second condition, $y^T (Ax - b) = 0$, is derived from the equality $y^T Ax = y^T b$. The vector $w = Ax - b$ represents the slack variables, and the equation $y^T w = 0$ implies that for each constraint i, either the dual variable $y_i = 0$ or the corresponding slack variable $w_i = 0$ (or both). This is expressed as $y_i w_i = 0$. The red underlines and boxes highlight the key equations and conclusions. The context is the development of optimality conditions in linear programming, which are crucial for understanding the relationship between primal and dual solutions and for developing efficient solution algorithms.

Figures

(a)

Slide 3

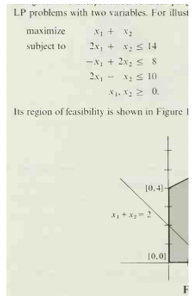
Content

THE GRAPHIC METHOD [C] p.260
The geometric interpretation of linear programming leads to a geometric procedure for solving LP problems with two variables. For illustration, let us consider the problem
maximize $\mathrm{x}_1 + x_2$
subject to $2x_1 + x_2 \leq 14$
$-x_1 + 2x_2 \leq 8$
$2x_1 - x_2 \leq 10$
$x_1, x_2 \geq 0.$
Its region of feasibility is shown in Figure 17.7, together with the line representing all the points

Description

Slide 3 of 38 introduces the graphic method for solving linear programming problems. It presents a linear programming problem with two variables, x_1 and x_2, aiming to maximize the objective function $x_1 + x_2$ subject to three inequality constraints and non-negativity constraints. The slide then explains that the feasible region of this problem is graphically represented in Figure 17.7. Figure 17.7 shows a shaded pentagon representing the feasible region, with its vertices labeled by their coordinates: (0, 0), (0, 4), (4, 6), (5, 0). A line representing the equation $x_1 + x_2 = 2$ is also plotted, intersecting the feasible region. The figure's purpose is to illustrate the geometric interpretation of linear programming, paving the way for a graphical solution method. The reference "[C] p.260" likely indicates a citation to a textbook or other source where this method is further explained. The context is the introduction of a geometric approach to solving linear programming problems, contrasting with the algebraic methods discussed in previous slides.

Figures



LP problems with two variables. For illust

maximize $x_1 + x_2$

subject to $2x_1 + x_2 \le 14$

$-x_1 + 2x_2 \le 8$

$2x_1 - x_2 \le 10$

$x_1, x_2 \ge 0.$

Its region of feasibility is shown in Figure 1

(a) A shaded pentagonal region representing the feasible region of a linear programming problem with two variables (x_1 and x_2). The vertices of the pentagon are labeled with their coordinates: [0, 0], [0, 4], [4, 6], [5, 0], and [0, 0]. A line representing $x_1 + x_2 = 2$ intersects the feasible region. The coordinates of the vertices are explicitly marked on the graph. The graph is labeled as "Figure 17.7".

Slide 4

Content

The geometric interpretation of linear programming leads to a geometric procedure for solving LP problems with two variables. For illustration, let us consider the problem

maximize x$_1 + x_2$

subject to $2x_1 + x_2 \leq 14$

$-x_1 + 2x_2 \leq 8$

$2x_1 - x_2 \leq 10$

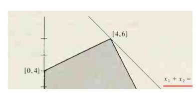
$x_1, x_2 \geq 0.$

Its region of feasibility is shown in Figure 17.8, together with the line representing all the points

Description

Slide 4 of 38 continues the explanation of the graphic method for solving linear programming problems. It shows a graphical representation (Figure 17.8) of the feasible region for a linear programming problem with two variables (x_1 and x_2). The objective function is to maximize $x_1 + x_2$, subject to the constraints $2x_1 + x_2 \leq 14$, $-x_1 + 2x_2 \leq 8$, $2x_1 - x_2 \leq 10$, and $x_1, x_2 \geq 0$. The feasible region is a pentagon with vertices at (0, 0), (0, 4), (4, 6), (5, 0), and (6,2). A line representing the equation $x_1 + x_2 = 10$ is drawn, intersecting the feasible region. This figure illustrates how the optimal solution can be found graphically by identifying the point in the feasible region that maximizes the objective function. The context is the continuation of the geometric approach to solving linear programming problems, building upon the previous slide's introduction of the graphic method.

Figures



(a) A shaded pentagonal region representing the feasible region of a linear programming problem with two variables (x_1 and x_2). The vertices of the pentagon are labeled with their coordinates: [0, 0], [0, 4], [4, 6], [5, 0], and [0, 0]. A line representing $x_1 + x_2 = 10$ intersects the feasible region. The coordinates of the vertices are explicitly marked on the graph. The graph is labeled as "Figure 17.8".

Slide 5

Content

$$\max \quad x_1 + x_2$$

$$(P) \text{ s.t. } \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$$\begin{cases} y_1(2x_1 + x_2) \leq 14y_1 \\ y_2(-x_1 + 2x_2) \leq 8y_2 \\ y_3(2x_1 - x_2) \leq 10y_3 \end{cases}$$

$$(2y_1 - y_2 + 2y_3)x_1 + (y_1 + 2y_2 - y_3)x_2 \leq 14y_1 + 8y_2 + 10y_3$$

Description

Slide 5 of 38 presents a primal linear programming problem (P) and its corresponding dual problem. The primal problem aims to maximize the objective function $x_1 + x_2$ subject to three inequality constraints and non-negativity constraints on x_1 and x_2. The dual problem is derived by introducing dual variables y_1, y_2, y_3 for each constraint in the primal problem. The dual constraints are obtained by multiplying each primal constraint by its corresponding dual variable and summing them up. The final inequality shows the dual objective function as a linear combination of the dual variables, derived from the dual constraints. The red arrow indicates the transformation from the primal problem to its dual. The context is the derivation of the dual problem from a given primal linear programming problem, a fundamental concept in linear programming theory.

Figures

(a)

Slide 6

Content

$$\max \quad \mathrm{x}_1 + x_2$$

$$(P)\ \text{s.t.} \quad \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$$(D)\min \quad 14y_1 + 8y_2 + 10y_3$$

$$\text{s.t.} \quad \begin{cases} 2y_1 - y_2 + 2y_3 \geq 1 \\ y_1 + 2y_2 - y_3 \geq 1 \\ y_1, y_2, y_3 \geq 0 \end{cases}$$

Description

Slide 6 of 38 presents a primal linear programming problem (P) and its corresponding dual linear programming problem (D). The primal problem (P) is a maximization problem with an objective function of $x_1 + x_2$ subject to three inequality constraints and non-negativity constraints on x_1 and x_2. The dual problem (D) is a minimization problem. Its objective function is $14y_1 + 8y_2 + 10y_3$, and it has two inequality constraints and non-negativity constraints on the dual variables y_1, y_2, and y_3. The problems are clearly labeled (P) and (D), respectively. The red curly braces highlight the constraints in both problems. The context is the presentation of a primal-dual pair of linear programming problems, derived from the primal problem introduced in the previous slides. This slide demonstrates the process of formulating the dual problem from a given primal problem, a key concept in linear programming theory and duality theory.

Figures

(a)

Slide 7

Content

$$\max \quad \mathrm{x}_1 + x_2$$

$$(P)\ \text{s.t.} \quad \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$$x_1 = 4, \quad x_2 = 6$$

$$(D)\min \quad 14y_1 + 8y_2 + 10y_3$$

$$\text{s.t.} \quad \begin{cases} 2y_1 - y_2 + 2y_3 \geq 1 \\ y_1 + 2y_2 - y_3 \geq 1 \\ y_1, y_2, y_3 \geq 0 \end{cases}$$

Description

Slide 7 of 38 shows the primal (P) and dual (D) linear programming problems. The primal problem is a maximization problem with objective function $x_1 + x_2$ subject to three inequality constraints and non-negativity constraints on x_1 and x_2. A solution to the primal problem is given as $x_1 = 4$, $x_2 = 6$. The dual problem is a minimization problem with objective function $14y_1 + 8y_2 + 10y_3$ subject to two inequality constraints and non-negativity constraints on y_1, y_2, y_3. The problems are clearly labeled (P) and (D), respectively. The red curly braces highlight the constraints in both problems. The context is the presentation of a solved primal problem and its corresponding dual problem, illustrating the relationship between primal and dual solutions in linear programming. The solution to the primal problem is explicitly stated, suggesting that this slide follows the presentation of a method to solve the primal problem (e.g., graphical method or simplex method).

Figures

(a)

Slide 8

Content

max $\quad \mathrm{x}_1 + x_2 = 10$

$$(P)\text{ s.t.} \quad \begin{cases} 2x_1 + x_2 \leq 14 \\ -x_1 + 2x_2 \leq 8 \\ 2x_1 - x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases}$$

$x_1 = 4, \quad x_2 = 6$
$z_1 = 0, \quad z_2 = 0$
$w_1 = 0$
$w_2 = 0$
$w_3 > 0 \implies y_3 = 0$
$(D)\text{min} \quad 14y_1 + 8y_2 + 10y_3 = 10$

$$\text{s.t.} \quad \begin{cases} 2y_1 - y_2 + 2y_3 \geq 1 \\ y_1 + 2y_2 - y_3 \geq 1 \\ y_1, y_2, y_3 \geq 0 \end{cases}$$

$y_1 = \frac{3}{5}, \quad y_2 = \frac{1}{5}$

Description

Slide 8 of 38 presents the primal (P) and dual (D) linear programming problems, along with their solutions. The primal problem is a maximization problem with objective function $x_1 + x_2 = 10$, subject to three inequality constraints and non-negativity constraints on x_1 and x_2. The optimal solution to the primal problem is given as $x_1 = 4$ and $x_2 = 6$. Slack variables z_1 and z_2 are introduced, and their values at the optimum are shown as 0. The dual problem is a minimization problem with objective function $14y_1 + 8y_2 + 10y_3 = 10$, subject to two inequality constraints and non-negativity constraints on y_1, y_2, y_3. The optimal solution to the dual problem is given as $y_1 = \frac{3}{5}$ and $y_2 = \frac{1}{5}$. The dual variable y_3 is 0, indicated by the implication $w_3 > 0 \implies y_3 = 0$. The red arrows indicate the correspondence between primal constraints and dual variables. The context is the presentation of the solved primal and dual problems, highlighting the optimal solutions and the relationship between primal and dual variables. The slide demonstrates the application of the simplex method or a similar technique to solve both the primal and dual problems, and shows the complementary slackness conditions.

Figures

(a)

Slide 9

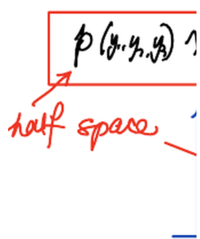
Content

Interpretation of

$(2y_1 - y_2 + 2y_3)x_1 + (y_1 + 2y_2 - y_3)x_2 \leq \underline{14y_1 + 8y_2 + 10y_3}$ $\boxed{p(y_1, y_2, y_3)x_1 + q(y_1, y_2, y_3)x_2 \leq r(y_1, y_2, y_3)}$ $\xrightarrow{\text{half space}}$ **Des**

Slide 9 of 38 interprets the constraints of the dual linear programming problem geometrically. The underlined expression shows a linear combination of x_1 and x_2 that is less than or equal to a linear combination of y_1, y_2, and y_3. This is then rewritten in a more compact form using functions p, q, and r of the dual variables y_1, y_2, and y_3. The boxed inequality represents a half-space in the x_1-x_2 plane. The accompanying graph illustrates this half-space, showing the region satisfying the inequality. The arrow points to the region representing the half-space. The context is the geometric interpretation of the dual problem's constraints, showing how each constraint defines a half-space in the primal variable space. This slide connects the algebraic representation of the dual problem to its geometric interpretation, a crucial concept in understanding the feasible region of the primal problem.

Figures



(a) A graph showing a half-space defined by a line in the x_1-x_2 plane. The line has a negative slope and intersects the x_1 axis at a positive value. An arrow points to the half-space below the line, indicating the region satisfying the inequality.

Slide 10

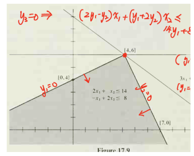
Content

$(2\mathrm{y}_1 - y_2)x_1 + (y_1 + 2y_2)x_2 \leq 14y_1 + 8y_2$
$y_3 = 0$
$y_1 = 0$
$5x_2 = 30$
$(y_1, y_2, y_3 = 2)$
$3x_1 + 4x_2 = 36$
$(y_1 = 2, y_2 = 1)$

Description

Slide 10 of 38 shows a graphical representation of the feasible region for a linear programming problem. The feasible region is a polygon in the x_1-x_2 plane, defined by three inequality constraints: $2x_1 + x_2 \leq 14$, $-x_1 + 2x_2 \leq 8$, and implicitly $2x_1 - x_2 \leq 10$. The vertices of the feasible region are clearly marked as (0, 4), (4, 6), and (7, 0). The optimal solution is highlighted at (4, 6). The slide also shows annotations relating the constraints to the dual variables (y_1, y_2, y_3), illustrating the connection between the primal and dual problems. Specifically, it shows how setting dual variables to zero ($y_3 = 0$, $y_1 = 0$) corresponds to specific constraints becoming active (or binding) in the primal problem. The equations $5x_2 = 30$ and $3x_1 + 4x_2 = 36$ are also shown, likely representing lines used in finding the optimal solution or illustrating the relationship between primal and dual solutions. The figure is labeled "Figure 17.9". The context is the geometric solution of a linear programming problem and the interpretation of the dual variables in relation to the primal feasible region.

Figures



(a) A graph showing a shaded feasible region in the x_1-x_2 plane. The region is bounded by three lines corresponding to the constraints $2x_1 + x_2 \leq 14$, $-x_1 + 2x_2 \leq 8$, and $2x_1 - x_2 \leq 10$. The vertices of the feasible region are labeled as [0, 4], [4, 6], and [7, 0]. Arrows indicate the direction of the inequalities. The point [4, 6] is marked with a red dot. The graph is labeled as "Figure 17.9". Additional annotations are present, including expressions involving dual variables (y_1, y_2, y_3) and equations relating primal and dual variables.

Slide 11

Content

Sensitivity Analysis (what if c, b change)

(P) max $\zeta = c^T x$

s.t. $\begin{cases} Ax \leq b \\ x \geq 0 \end{cases}$

(D) min $\xi = b^T y$

s.t. $\begin{cases} A^T y \geq c \\ y \geq 0 \end{cases}$

Description

Slide 11 of 38 introduces sensitivity analysis in linear programming. The slide presents the primal (P) and dual (D) linear programming problems. The primal problem is a maximization problem with objective function $\zeta = c^T x$, subject to the constraints $Ax \leq b$ and $x \geq 0$. The dual problem is a minimization problem with objective function $\xi = b^T y$, subject to the constraints $A^T y \geq c$ and $y \geq 0$. The title "Sensitivity Analysis (what if c, b change)" indicates that the slide will discuss how changes in the cost vector c and the resource vector b affect the optimal solutions of both the primal and dual problems. The underlined title emphasizes the focus of the analysis. The notation is standard for linear programming, with A representing the constraint matrix, x and y representing the primal and dual variables respectively, and b and c representing the resource vector and cost vector respectively. The context is the introduction of sensitivity analysis, a crucial topic in linear programming that deals with the robustness of optimal solutions to changes in problem parameters.

Figures

(a)

Slide 12

Content

Sensitivity Analysis (what if c, b change)

(P) max $\zeta = c^T x$

s.t. $\begin{cases} Ax \leq b \\ x \geq 0 \end{cases}$

(D) min $\xi = b^T y$

s.t. $\begin{cases} A^T y \geq c \\ y \geq 0 \end{cases}$

$\zeta(x_B, x_N) = c_B^T B^{-1} b - (B^{-1}N)^T (c_B - c_N) x_N$

$x_B = B^{-1}b - (B^{-1}N)x_N$

at opt. $x_N = 0, \quad x_B = B^{-1}b \geq 0$

$(B^{-1}N)^T (c_B - c_N) \geq 0$

$\zeta^* = c_B^T x_B = c^T x^*$

Description

Slide 12 of 38 delves into sensitivity analysis in linear programming, focusing on how changes in the cost vector c and the resource vector b impact the optimal solution. It begins by restating the primal (P) and dual (D) linear programs. Then, it presents equations that describe the optimal solution's sensitivity to changes in b and c. The equation $\zeta(x_B, x_N) = c_B^T B^{-1} b - (B^{-1}N)^T (c_B - c_N) x_N$ shows how the objective function value ζ depends on the basic (x_B) and non-basic (x_N) variables. The equation $x_B = B^{-1}b - (B^{-1}N)x_N$ expresses the basic variables in terms of the non-basic variables and the resource vector b. At the optimal solution, $x_N = 0$, simplifying these equations. The condition $(B^{-1}N)^T (c_B - c_N) \geq 0$ is a necessary condition for optimality. Finally, $\zeta^* = c_B^T x_B = c^T x^*$ confirms that the optimal objective function value is the same whether calculated using basic or all variables. The notation uses standard linear programming conventions: A is the constraint matrix, B is the basis matrix, N is the non-basis matrix, c_B and c_N are the cost vectors for basic and non-basic variables, and x_B and x_N are the basic and non-basic variables. The underlined title highlights the core topic: sensitivity analysis. The context is the mathematical formulation of sensitivity analysis, providing the tools to assess how changes in problem parameters affect the optimal solution.

Figures

(a)

Slide 13

Content

Sensitivity Analysis (what if c, b change)

(P) max $\zeta = c^T x$

s.t. $\begin{cases} Ax \leq b \\ x \geq 0 \end{cases}$

(D) min $\xi = b^T y$

s.t. $\begin{cases} A^T y \geq c \\ y \geq 0 \end{cases}$

$\xi(z_B, z_N) = -c_B^T B^{-1} b - (B^{-1}b)^T z_B$
$z_N = (B^{-1}N)^T (c_B - c_N) + (B^{-1}N)^T z_B$
at opt. $z_B^* = 0, \quad z_N = (B^{-1}N)^T (c_B - c_N) \geq 0$
$B^{-1} b \geq 0$
$\xi^* = c_B^T B^{-1} b = b^T y^* \quad (y = (B^{-1})^T c_B)$

Description

Slide 13 of 38 continues the discussion on sensitivity analysis in linear programming, focusing on the impact of changes in the cost vector c and the resource vector b on the optimal solution. The slide begins by restating the primal (P) and dual (D) linear programs. It then introduces equations that describe the optimal solution's sensitivity. The equation $\xi(z_B, z_N) = -c_B^T B^{-1} b - (B^{-1}b)^T z_B$ expresses the dual objective function value ξ in terms of basic (z_B) and non-basic (z_N) dual variables. The equation $z_N = (B^{-1}N)^T (c_B - c_N) + (B^{-1}N)^T z_B$ shows how the non-basic dual variables depend on the basic dual variables, the cost vectors, and the basis and non-basis matrices. At the optimal solution, $z_B^* = 0$, simplifying these equations. The condition $(B^{-1}N)^T (c_B - c_N) \geq 0$ is a necessary condition for optimality. The condition $B^{-1}b \geq 0$ ensures primal feasibility. Finally, $\xi^* = c_B^T B^{-1} b = b^T y^*$ shows the equality of the optimal objective function values for the primal and dual problems, with the expression $y = (B^{-1})^T c_B$ providing the optimal dual solution. The notation is consistent with standard linear programming conventions, using A, B, N, c_B, c_N, z_B, and z_N to represent the constraint matrix, basis matrix, non-basis matrix, cost vectors for basic and non-basic variables, and basic and non-basic dual variables, respectively. The underlined title emphasizes the focus on sensitivity analysis. The context is the mathematical derivation of sensitivity analysis results, providing the tools to analyze how changes in problem parameters affect the optimal solution.

Figures

(a)

Slide 14

Content

Sensitivity Analysis (what if c, b change)

(P) max $\zeta = c^T x$

s.t. $\begin{cases} Ax \leq b \\ x \geq 0 \end{cases}$

(D) min $\xi = b^T y$

s.t. $\begin{cases} A^T y \geq c \\ y \geq 0 \end{cases}$

at Opt. $\zeta^* = c^T x^* = b^T y^* = \xi^*$

$\downarrow$ if only c changes

$\Delta\zeta^* = (\Delta c)^T x^*$

Description

Slide 14 of 38 focuses on sensitivity analysis in linear programming, specifically examining the effects of changes in the cost vector c and the resource vector b on the optimal solution. The slide begins by presenting the primal (P) and dual (D) linear programming problems. The primal problem maximizes $\zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$, while the dual problem minimizes $\xi = b^T y$ subject to $A^T y \geq c$ and $y \geq 0$. At the optimum, the primal and dual objective functions are equal: $\zeta^* = c^T x^* = b^T y^* = \xi^*$. The core of the slide is the analysis of changes in the optimal objective function value when only the cost vector c changes. The equation $\Delta\zeta^* = (\Delta c)^T x^*$ shows that the change in the optimal objective function value ($\Delta\zeta^*$) is equal to the transpose of the change in the cost vector (Δc) multiplied by the optimal solution vector (x^*). This equation provides a direct way to calculate the impact of changes in the cost vector on the optimal objective function value, assuming the optimal solution remains feasible. The downward-pointing arrow emphasizes that this equation applies specifically when only c changes. The context is sensitivity analysis within the broader topic of linear programming, providing a formula for quickly assessing the impact of cost changes on the optimal objective function value.

Figures

(a)

Slide 15

Content

Sensitivity Analysis (what if c, b change)

(P) max $\zeta = c^T x$

s.t. $\begin{cases} Ax \leq b \\ x \geq 0 \end{cases}$

(D) min $\xi = b^T y$

s.t. $\begin{cases} A^T y \geq c \\ y \geq 0 \end{cases}$

at Opt. $\zeta^* = c^T x^* = b^T y^* = \xi^*$

$\downarrow$ if only b changes

$\Delta\zeta^*(= \Delta\xi^*) = (\Delta b)^T y^*$

y^* : shadow prices

Description

Slide 15 of 38 continues the sensitivity analysis in linear programming, focusing on the effects of changes in the cost vector c and the resource vector b. It presents the primal (P) and dual (D) linear programs, showing that at the optimum, the primal and dual objective functions are equal: $\zeta^* = c^T x^* = b^T y^* = \xi^*$. The key part of the slide analyzes the change in the optimal objective function value when only the resource vector b changes. The equation $\Delta\zeta^*(= \Delta\xi^*) = (\Delta b)^T y^*$ shows that the change in the optimal objective function value ($\Delta\zeta^*$, which is equal to the change in the dual optimal objective function value $\Delta\xi^*$) is the transpose of the change in the resource vector (Δb) multiplied by the optimal dual solution vector (y^*). This equation provides a direct way to calculate the impact of changes in the resource vector on the optimal objective function value, assuming the optimal solution remains feasible. The downward-pointing arrow indicates that this equation applies when only b changes. The optimal dual variables y^* are labeled as "shadow prices," indicating their economic interpretation as the marginal value of an additional unit of each resource. The context is sensitivity analysis in linear programming, providing a formula to assess the impact of resource changes on the optimal objective function value and introducing the concept of shadow prices.

Figures

(a)

Slide 16

Content

4. Consider the following linear programming problem:
maximize $\mathrm{x}_1 + 2x_2$

$$\text{subject to } \begin{cases} x_1 + x_2 \leq 5 \\ x_1 \leq 4 \\ x_2 \leq 3 \\ x_1, x_2 \geq 0 \end{cases}$$

(a) Solve the above problem and give also the dictionary at the optimal point.
(b) Suppose the constraint $x_1 + x_2 \leq 5$ is changed to $x_1 + x_2 \leq p$. Find the range of p such that the dictionary you have found in (a) remains optimal.
(c) Suppose the objective function $x_1 + 2x_2$ is changed to $x_1 + qx_2$. Find the range of q such that the dictionary you have found in (a) remains optimal.

Description

Slide 16 of 38 presents a linear programming problem and three follow-up questions related to sensitivity analysis. The problem is to maximize $x_1 + 2x_2$ subject to the constraints $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, and $x_1, x_2 \geq 0$. Part (a) asks to solve this linear program and provide the optimal dictionary (simplex tableau). Parts (b) and (c) are sensitivity analyses. Part (b) investigates how changing the right-hand side of the constraint $x_1 + x_2 \leq 5$ to p affects the optimality of the dictionary found in part (a). It asks for the range of p for which the optimal dictionary remains unchanged. Part (c) explores the impact of changing the objective function's coefficient of x_2 from 2 to q. It seeks the range of q that preserves the optimality of the dictionary from part (a). The context is sensitivity analysis in linear programming, focusing on determining the range of parameter changes that maintain the optimality of a given solution.

Figures

(a)

Slide 17

Content

4. Consider the following linear programming problem:
maximize $\mathrm{x}_1 + 2x_2$

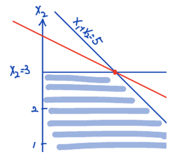
subject to $\begin{cases} x_1 + x_2 \leq 5 \\ x_1 \leq 4 \\ x_2 \leq 3 \\ x_1, x_2 \geq 0 \end{cases}$

(a) Solve the above problem and give also the dictionary at the optimal point.
(b) Suppose the constraint $x_1 + x_2 \leq 5$ is changed to $x_1 + x_2 \leq p$. Find the range of p such that the dictionary you have found in (a) remains optimal.
(c) Suppose the objective function $x_1 + 2x_2$ is changed to $x_1 + qx_2$. Find the range of q such that the dictionary you have found in (a) remains optimal.

Description

Slide 17 of 38 presents a linear programming problem and three questions related to sensitivity analysis. The problem is to maximize $x_1 + 2x_2$ subject to $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, $x_1, x_2 \geq 0$. Part (a) requires solving this linear program and providing the optimal dictionary (simplex tableau). Parts (b) and (c) involve sensitivity analysis. Part (b) asks to find the range of p such that the optimal dictionary remains optimal if the constraint $x_1 + x_2 \leq 5$ is changed to $x_1 + x_2 \leq p$. Part (c) asks to find the range of q such that the optimal dictionary remains optimal if the objective function is changed from $x_1 + 2x_2$ to $x_1 + qx_2$. The figure graphically represents the feasible region defined by the constraints. The optimal solution is visually identified at the intersection of $x_1 + x_2 = 5$ and $x_2 = 3$, which is $(2, 3)$. The objective function line $x_1 + 2x_2 = 8$ is also shown. The context is sensitivity analysis in linear programming, focusing on how changes in constraints and objective function coefficients affect the optimal solution.

Figures



(a) A graph showing the feasible region of the linear program, with constraints $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, $x_1 \geq 0$, $x_2 \geq 0$. The optimal solution is marked with a red dot at the intersection of $x_1 + x_2 = 5$ and $x_2 = 3$, which is $(2, 3)$. The objective function $x_1 + 2x_2 = 8$ is also plotted as a red line. The feasible region is shaded in light blue. The axes are labeled x_1 and x_2. The lines representing the constraints are labeled accordingly.

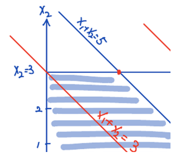
Slide 18

Content

Description

Slide 18 of 38 graphically illustrates the sensitivity analysis of a linear program. The feasible region is defined by the constraints $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, $x_1 \geq 0$, and $x_2 \geq 0$. The graph shows multiple parallel lines representing different objective functions of the form $x_1 + qx_2 = c$, where q is the coefficient of x_2 and c is a constant. The lines illustrate how the optimal solution changes as the objective function changes (by varying q). The red line, labeled $x_1 + x_2 = 3$, is one such objective function. The optimal solution for the original objective function ($x_1 + 2x_2$) is at the intersection of $x_1 + x_2 = 5$ and $x_2 = 3$, which is $(2, 3)$. The arrows indicate the direction of increasing objective function value. The graph visually demonstrates the concept of sensitivity analysis by showing how the optimal solution changes as the objective function's slope changes. The context is sensitivity analysis in linear programming, specifically visualizing the impact of changes in the objective function's coefficients on the optimal solution.

Figures



(a) A graph showing the feasible region of a linear program. The constraints are $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, $x_1 \geq 0$, $x_2 \geq 0$. The feasible region is shaded in light blue. Several parallel lines representing different objective functions ($x_1 + qx_2 = c$, for various values of c and q) are drawn within the feasible region. One line, $x_1 + x_2 = 3$, is highlighted in red. The optimal solution is at the intersection of $x_1 + x_2 = 5$ and $x_2 = 3$, which is $(2, 3)$. Arrows indicate the direction of improvement of the objective function. The axes are labeled x_1 and x_2. The lines representing the constraints are labeled accordingly.

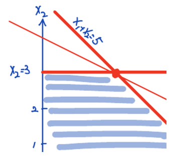
Slide 19

Content

Description

Slide 19 of 38 graphically illustrates the sensitivity analysis of a linear program. The feasible region is defined by the constraints $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, $x_1 \geq 0$, and $x_2 \geq 0$. The graph shows the feasible region shaded with parallel lines. The optimal solution for the objective function $x_1 + 2x_2$ is at (2, 3). The graph also shows parallel lines representing different objective functions of the form $x_1 + qx_2 = c$, where q represents the coefficient of x_2 and c is a constant. The lines illustrate how the optimal solution changes as the objective function changes (by varying q). The red line $x_1 + 2x_2 = 8$ represents the original objective function. The labels $q = 1$ and $q = \infty$ indicate different objective function slopes. Arrows show the direction of increasing objective function value. The graph visually demonstrates the concept of sensitivity analysis by showing how the optimal solution changes as the objective function's slope changes. The context is sensitivity analysis in linear programming, specifically visualizing the impact of changes in the objective function's coefficients on the optimal solution.

Figures



(a) A graph showing the feasible region of a linear program. The constraints are $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, $x_1 \geq 0$, $x_2 \geq 0$. The feasible region is shaded with light blue parallel lines. A red line representing the objective function $x_1 + 2x_2 = 8$ passes through the optimal solution point (2, 3). Another red line, representing $x_1 + x_2 = 5$, is also shown. The axes are labeled x_1 and x_2. The lines representing the constraints are labeled accordingly. Arrows indicate the direction of improvement of the objective function. The labels $x_2 = 3$, $x_1 = 4$, $q = 1$, and $q = \infty$ are also present on the graph.

Slide 20

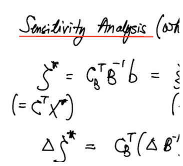
Content

Sensitivity Analysis (what if c , b changes)
S$^* = c_B^T B^{-1} b = S^*$
$(= c^T x^*)$ $(= b^T y^*)$
$\Delta S^* = c_B^T (\Delta B^{-1}) b$
$\underline{\Delta(B^{-1}) = -B^{-1}(\Delta B) B^{-1}}$
$\Delta S^* = -c_B^T B^{-1} (\Delta B) B^{-1} b$

Description

Slide 20 of 38 presents formulas for sensitivity analysis in linear programming, specifically focusing on how changes in the right-hand side vector (b) of the constraints affect the optimal objective function value (S^*). The slide starts by defining the optimal objective function value as $S^* = c_B^T B^{-1} b$, where c_B is the cost vector corresponding to the basic variables, B is the basis matrix, and b is the right-hand side vector. The change in the optimal objective function value, ΔS^*, is then expressed as $c_B^T (\Delta B^{-1}) b$, where ΔB^{-1} represents the change in the inverse of the basis matrix. A key formula is derived: $\Delta(B^{-1}) = -B^{-1}(\Delta B) B^{-1}$, which provides a way to calculate the change in the inverse of the basis matrix given a change in the basis matrix itself. Finally, this is substituted to express ΔS^* as $-c_B^T B^{-1} (\Delta B) B^{-1} b$. The crossed-out 'c' in the title indicates that the analysis focuses solely on changes in b, not c. The context is sensitivity analysis within the broader topic of linear programming, specifically dealing with post-optimality analysis to understand the impact of changes in problem data on the optimal solution.

Figures



(a) Handwritten notes on sensitivity analysis in linear programming. The notes include equations for calculating the change in the optimal objective function value (ΔS^*) when changes occur in the constraint matrix (B). The equations involve matrix operations, including matrix inversion (B^{-1}) and matrix multiplication. A crossed-out 'c' indicates that this analysis focuses on changes in 'b' rather than 'c'.

Slide 21

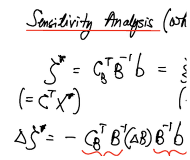
Content

Sensitivity Analysis (what if c, b changes) $S^* = c_B^T B^{-1} b = S^*$ $(= c^T x^*)$ $(= b^T y^*)$ $\Delta S^* = -c_B^T B^{-1} (\Delta B) B^{-1} b$ $\underbrace{(y^*)^T}_{c_B^T B^{-1}} \underbrace{x_B^*}_{B^{-1} b} = -(y^*)^T (\Delta B) x_B^* = -\langle (\Delta B) x_B^*, y^* \rangle$

Description

Slide 21 of 38 continues the sensitivity analysis from the previous slide, focusing on the impact of changes in the constraint matrix (B) on the optimal objective function value (S^*). The slide begins by restating the optimal objective function value as $S^* = c_B^T B^{-1} b$, where c_B is the cost vector of basic variables, B is the basis matrix, and b is the right-hand side vector. It then presents the formula for the change in the optimal objective function value, $\Delta S^* = -c_B^T B^{-1} (\Delta B) B^{-1} b$, where ΔB represents the change in the basis matrix. The equation is then rewritten to highlight the terms $c_B^T B^{-1}$ and $B^{-1} b$, which are identified as $(y^*)^T$ and x_B^* respectively, where y^* is the optimal dual solution and x_B^* is the optimal primal basic solution. Finally, the equation is expressed concisely using inner product notation: $\Delta S^* = -\langle (\Delta B) x_B^*, y^* \rangle$. The crossed-out 'c' in the title reinforces that the analysis focuses on changes in b, not c. The context is sensitivity analysis in linear programming, specifically showing how changes in the constraint matrix affect the optimal solution and objective function value. The use of inner product notation simplifies the expression and emphasizes the relationship between the primal and dual solutions.

Figures



(a) Handwritten notes on sensitivity analysis in linear programming. The notes include equations for calculating the change in the optimal objective function value (ΔS^*) when changes occur in the constraint matrix (B). The equations involve matrix operations, including matrix inversion (B^{-1}) and matrix multiplication. A crossed-out 'c' indicates that this analysis focuses on changes in 'b' rather than 'c'. The final equation is expressed using inner product notation.

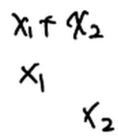
Slide 22

Content

Description

Slide 22 of 38 presents a linear programming example problem from a midterm exam. The problem is to maximize the objective function $x_1 + 2x_2$ subject to the constraints $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, and $x_1, x_2 \geq 0$. The problem is then formulated in matrix form, where A represents the constraint matrix and N represents the slack variables. The matrix equation shows the relationship between the decision variables (x_1, x_2), slack variables (w_1, w_2, w_3), and the right-hand side vector. The context is linear programming, specifically showing how a linear program can be represented in matrix notation, which is a crucial step for solving it using algorithms like the simplex method. The problem is labeled as "Midterm 4," indicating its origin from a past exam.

Figures



(a) Matrix representation of the linear program.

Slide 23

Content

Ex (midterm #4)
max $\mathrm{x}_1 + 2x_2$
s.t.
$x_1 + x_2 \leq 5$
$x_1 \leq 4$
$x_2 \leq 3$
$x_1, x_2 \geq 0$

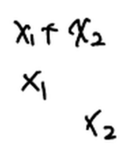
$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 3 \end{bmatrix}$$

$A \quad N$
at Opt:
$x_1^* = 2, \quad x_2^* = 3$
$w_1^* = 0$
$w_2^* > 0$
$w_3^* = 0$

Description

Slide 23 of 38 presents a linear programming example problem from a midterm exam. The problem is to maximize the objective function $x_1 + 2x_2$ subject to the constraints $x_1 + x_2 \leq 5$, $x_1 \leq 4$, $x_2 \leq 3$, and $x_1, x_2 \geq 0$. The problem is then formulated in matrix form, where A represents the constraint matrix and N represents the slack variables. The matrix equation shows the relationship between the decision variables (x_1, x_2), slack variables (w_1, w_2, w_3), and the right-hand side vector. The optimal solution is also given: $x_1^* = 2$, $x_2^* = 3$, with $w_1^* = 0$, $w_2^* > 0$, and $w_3^* = 0$. The context is linear programming, specifically showing how a linear program can be represented in matrix notation, which is a crucial step for solving it using algorithms like the simplex method. The problem is labeled as "Midterm 4," indicating its origin from a past exam. The optimal solution indicates which constraints are binding (those with zero slack) and which are not.

Figures



(a) Matrix representation of the linear program.

Slide 24

Content

$$\mathrm{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}; \quad \mathrm{B} = \{1, 2, 4\}$$

$$\mathrm{x}_B^* = B^{-1}b$$

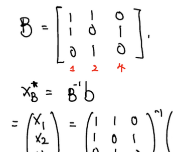
$$= \begin{pmatrix} x_1 \\ x_2 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \leftarrow \begin{matrix} x_1^* \\ x_2^* \\ w_2^* \end{matrix}$$

Description

Slide 24 of 38 shows the calculation of the optimal basic solution (x_B^*) for a linear programming problem. The basis matrix B, consisting of columns 1, 2, and 4 from the original constraint matrix (as indicated by $\mathcal{B} = \{1, 2, 4\}$), is shown. Its inverse, B^{-1}, is calculated and used to find the optimal solution $x_B^* = B^{-1}b$, where b is the right-hand side vector $\begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}$. The resulting optimal solution is $x_1^* = 2$, $x_2^* = 3$, and $w_2^* = 2$. The arrows indicate the correspondence between the elements of the resulting vector and the variables x_1, x_2, and w_2. The context is solving a linear program using the simplex method, where this calculation is a crucial step in determining the optimal values of the decision variables and slack variables.

Figures



(a) Handwritten notes showing the calculation of the optimal basic solution x_B^* in a linear programming problem. The notes include a basis matrix B, its inverse B^{-1}, and the right-hand side vector b. The optimal solution is obtained by multiplying the inverse of the basis matrix by the right-hand side vector. The indices 1, 2, and 4 indicate the columns of the original constraint matrix that form the basis.

Slide 25

Content

$$\mathrm{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}; \quad \mathrm{B} = \{1, 2, 4\}$$

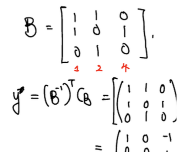
$$\mathrm{y}^T = (B^{-1})^T c_B = \begin{bmatrix} (1 & 1 & 0) \\ (0 & 1 & 0) \\ (1 & 0 & 1) \end{bmatrix}^{-1} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \leftarrow \begin{matrix} y_1 \\ y_2 \\ y_3 \end{matrix}$$

Description

Slide 25 of 38 shows the calculation of the optimal dual solution (y^T) for a linear programming problem. The basis matrix B, consisting of columns 1, 2, and 4 from the original constraint matrix (as indicated by $\mathcal{B} = \{1, 2, 4\}$), is shown. Its inverse, B^{-1}, is calculated and used to find the optimal dual solution $y^T = (B^{-1})^T c_B$, where c_B is the cost vector $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ corresponding to the basic variables. The resulting optimal dual solution is $y_1 = 1$, $y_2 = 0$, and $y_3 = 1$. The arrows indicate the correspondence between the elements of the resulting vector and the dual variables y_1, y_2, and y_3. The context is solving a linear program using the simplex method, where this calculation is a crucial step in determining the optimal values of the dual variables, which provide information about the shadow prices of the constraints.

Figures



(a) Handwritten notes showing the calculation of the optimal dual solution y^T in a linear programming problem. The notes include a basis matrix B, its inverse B^{-1}, and the cost vector c_B. The optimal dual solution is obtained by multiplying the transpose of the inverse of the basis matrix by the cost vector. The indices 1, 2, and 4 indicate the columns of the original constraint matrix that form the basis.

Slide 26

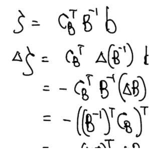
Content

$\zeta = c_B^T B^{-1} b$
$\Delta\zeta = c_B^T \Delta(B^{-1}) b$
$= -c_B^T B^{-1} (\Delta B) B^{-1} b$
$= -((B^{-1})^T c_B)^T \Delta B (B^{-1} b)$
$= -(y^*)^T \Delta B x_B^*$
$= -(1 \quad 0 \quad 0)(\Delta B) \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$

Description

Slide 26 of 38 demonstrates the calculation of the change in the objective function value ($\Delta\zeta$) resulting from a change in the basis matrix (ΔB) in a linear programming problem. The derivation starts with the expression for the objective function value ($\zeta = c_B^T B^{-1} b$) and then calculates the change ($\Delta\zeta$) using the formula $\Delta\zeta = c_B^T \Delta(B^{-1}) b$. This is then manipulated using matrix algebra, ultimately leading to the expression $\Delta\zeta = -(y^*)^T \Delta B x_B^*$. This shows that the change in the objective function value is a function of the change in the basis matrix (ΔB), the optimal dual solution (y^*), and the optimal primal solution (x_B^*). The final line shows a numerical example, substituting a specific value for y^* and x_B^* obtained from previous calculations (likely from slides 24 and 25). The context is sensitivity analysis within the simplex method for linear programming, where understanding how changes in the problem data affect the optimal solution is crucial.

Figures



(a) Handwritten notes showing the calculation of the change in the objective function value ($\Delta\zeta$) in a linear programming problem. The notes involve matrix operations using the cost vector (c_B), the inverse of the basis matrix (B^{-1}), the change in the basis matrix (ΔB), and the optimal solution vector (x_B^*). The optimal dual solution vector (y^*) is also used. The final expression shows a specific numerical calculation using a particular value for y^* and x_B^*.

Slide 27

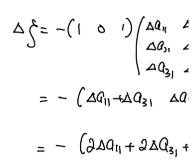
Content

$$
\begin{aligned}
\Delta\zeta &= -\begin{pmatrix}1 & 0 & 1\end{pmatrix}\begin{pmatrix}\Delta a_{11} & \Delta a_{12} & 0\\ \Delta a_{21} & \Delta a_{22} & 0\\ \Delta a_{31} & \Delta a_{32} & 0\end{pmatrix}\begin{pmatrix}2\\3\\2\end{pmatrix}\\
&= -\begin{pmatrix}\Delta a_{11}+\Delta a_{31} & \Delta a_{12}+\Delta a_{32} & 0\end{pmatrix}\begin{pmatrix}2\\3\\2\end{pmatrix}\\
&= -(2\Delta a_{11}+2\Delta a_{31}+3\Delta a_{12}+3\Delta a_{32})
\end{aligned}
$$

Description

Slide 27 of 38 shows the calculation of the change in the objective function value ($\Delta\zeta$) due to changes in the constraint matrix. The calculation uses matrix multiplication: a row vector (1 0 1), which appears to be derived from the optimal dual solution (from slide 25), is multiplied by a matrix representing changes in the constraint matrix (ΔA), and then by a column vector (2 3 2) representing the optimal primal solution (from slide 24). The matrix ΔA contains elements Δa_{ij} representing changes in the coefficients of the constraint matrix. The result of the matrix multiplication is a scalar value representing the total change in the objective function value, $\Delta\zeta$. This calculation is a key part of sensitivity analysis in linear programming, allowing one to assess the impact of changes in the problem data on the optimal solution and objective function value. The context is sensitivity analysis within the simplex method for linear programming.

Figures



(a) Handwritten notes showing a matrix calculation to determine the change in the objective function value ($\Delta\zeta$). The calculation involves multiplying a row vector, a matrix, and a column vector. The matrix represents changes in the constraint matrix (Δa_{ij}), and the vectors represent the optimal primal and dual solutions. The final result is a scalar value representing the total change in the objective function.

Slide 28

Content

$\mathrm{x}_B^* = B^{-1}b$
$\Delta x_B^* = \Delta(B^{-1})b$
$= -B^{-1}(\Delta B)B^{-1}b$
$y^* = (B^{-1})^T c_B$
$\Delta y^* = -(B^{-1}(\Delta B)B^{-1})^T c_B$
$= -(B^{-1})^T(\Delta B)^T(B^{-1})^T c_B$

Description

Slide 28 of 38 presents formulas for calculating the changes in the optimal primal and dual solutions (Δx_B^* and Δy^*) due to changes in the basis matrix (ΔB) within the context of sensitivity analysis in linear programming. The slide begins by defining the optimal primal basic solution as $x_B^* = B^{-1}b$, where B^{-1} is the inverse of the basis matrix and b is the right-hand side vector. The change in the optimal primal solution, Δx_B^*, is then derived using matrix calculus, resulting in $\Delta x_B^* = -B^{-1}(\Delta B)B^{-1}b$. Similarly, the optimal dual solution is defined as $y^* = (B^{-1})^T c_B$, where c_B is the cost vector associated with the basic variables. The change in the optimal dual solution, Δy^*, is derived as $\Delta y^* = -(B^{-1}(\Delta B)B^{-1})^T c_B$, which is further simplified to $\Delta y^* = -(B^{-1})^T(\Delta B)^T(B^{-1})^T c_B$. These formulas are crucial for sensitivity analysis, allowing one to quantify the impact of small changes in the constraint matrix on the optimal primal and dual solutions. The superscript * denotes optimality, and the subscript B denotes basic variables. The context is sensitivity analysis within the simplex method for linear programming.

Figures

$$x_B^* = B^{-1} b$$

$$\Delta x_B^* = \Delta(B^{-1}) b$$

$$= -B^{-1}(\Delta B) B^{-1}$$

$$y^* = (B^{-1})^T c_B$$

(a) Handwritten notes showing the calculation of changes in the optimal primal and dual solutions (Δx_B^* and Δy^*) in a linear programming problem. The notes involve matrix operations using the inverse of the basis matrix (B^{-1}), the change in the basis matrix (ΔB), the cost vector (c_B), and the right-hand side vector (b). The optimal primal and dual solutions (x_B^* and y^*) are also shown. The equations show how changes in the basis matrix affect the optimal solutions.

Slide 29

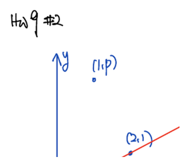
Content

y = $1_{\overline{2x}}$

Description

Slide 29 of 38 displays a hand-drawn graph depicting a linear equation, $y = \frac{1}{2}x$. The graph shows the line passing through the origin (0, 0) and the point (2, 1). Another point, approximately (4, 2), is also marked on the line, reinforcing the linear relationship. The axes are clearly labeled x and y. The title "Hw 9 2" suggests this is part of homework assignment 2 for week 9. The context of the course (Linear Programming) suggests this graph might illustrate a constraint or objective function in a simple linear programming problem. The simplicity of the graph and the equation suggest it's likely an introductory example used to illustrate basic concepts.

Figures



(a) A hand-drawn graph showing a line with the equation $y = \frac{1}{2}x$. The line passes through points (0, 0) and (2, 1). Another point (approximately (4, 2)) is also marked on the line. The axes are labeled x and y. The title "Hw 9 2" is written in the upper left corner. Points (0,0), (1,1/2), (2,1), and (4,2) are marked on the graph.

Slide 30

Content

y = ax + b $\left(a = \frac{1}{2}, b = 0\right)$

(P) min $x_1 + x_2 + x_3 + x_4$

$s.t.$ $|0 - (a(0) + b)| \leq x_1$

$|1 - (a(2) + b)| \leq x_2$

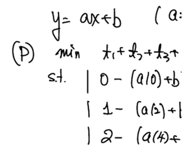
$|2 - (a(4) + b)| \leq x_3$

$|p - (a(i) + b)| \leq x_4$

Description

Slide 30 of 38 presents a linear programming problem (P) formulated using absolute values. The objective is to minimize the sum of four non-negative variables: $x_1 + x_2 + x_3 + x_4$. The constraints involve a linear function $y = \frac{1}{2}x$ (since $a = \frac{1}{2}$ and $b = 0$). The constraints are expressed as absolute value inequalities: $|y_i - (ax_i + b)| \leq x_i$, where y_i represents the constant values 0, 1, 2, and p, and x_i represents the corresponding variables x_1, x_2, x_3, x_4. This formulation likely aims to approximate the linear function using a piecewise linear model, where the variables x_i represent deviations from the linear function at specific points. The value of 'i' and 'p' are not specified on the slide. The context of the course (Linear Programming) suggests this is an example of how to model a problem with absolute values using linear programming techniques. The use of absolute values requires reformulation into a standard linear program, likely covered in subsequent slides.

Figures



(a) Handwritten notes showing a linear programming problem (P). The objective function is to minimize the sum of four variables (x_1, x_2, x_3, x_4). The constraints involve absolute values of differences between a linear function ($y = ax + b$ with $a = 1/2$ and $b = 0$) evaluated at different points (0, 2, 4, and i) and some constants (0, 1, 2, and p). Each absolute value is less than or equal to one of the variables to be minimized.

Slide 31

Content

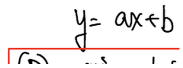
$$y = \text{ax} + \text{b} \quad \left(a = \frac{1}{2}, b = 0\right)$$

$$(P) \quad \min \quad x_1 + x_2 + x_3 + x_4$$
$$s.t. \quad -x_1 \leq -b \leq x_1$$
$$-x_2 \leq 1 - 2a - b \leq x_2$$
$$-x_3 \leq 2 - 4a - b \leq x_3$$
$$-x_4 \leq p - ai - b \leq x_4$$

Description

Slide 31 of 38 presents a linear programming problem (P) formulated to approximate a linear function using absolute values. The objective is to minimize the sum of four non-negative variables x_1, x_2, x_3, x_4. The constraints are expressed as double inequalities, which are equivalent to absolute value inequalities. Specifically, they represent the absolute difference between the linear function $y = \frac{1}{2}x$ (since $a = \frac{1}{2}$ and $b = 0$) evaluated at points 0, 2, 4, and an unspecified point 'i', and corresponding constant values 0, 1, 2, and p. Each absolute value is bounded above and below by the corresponding variable x_i, effectively minimizing the deviation from the linear function at those points. The values of 'i' and 'p' are undefined in the slide. This formulation likely aims to approximate the linear function using a piecewise linear model. The context of the course (Linear Programming) suggests this is an example of how to model a problem involving absolute values using linear programming techniques. The use of double inequalities is a common way to represent absolute value constraints in linear programming.

Figures



(a) Handwritten notes showing a linear programming problem (P). The objective function is to minimize the sum of four variables (x_1, x_2, x_3, x_4). The constraints involve inequalities with absolute values, where each absolute value is less than or equal to one of the variables to be minimized. The constraints are related to a linear function ($y = ax + b$ with $a = 1/2$ and $b = 0$) evaluated at different points (0, 2, 4, and i). The values of 'i' and 'p' are not specified.

Slide 32

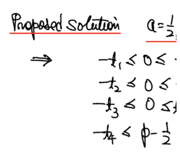
Content

Proposed Solution $a = \frac{1}{2, \quad b=0 \Longrightarrow -x_1 \leq 0 \leq x_1 \Longrightarrow x_1=0 - x_2 \leq 0 \leq x_2 \Longrightarrow x_2=0 - x_3 \leq 0 \leq x_3 \Longrightarrow x_3=0 - x_4 \leq p-\frac{i}{2} \leq x_4 \Longrightarrow x_4=\left|p-\frac{i}{2}\right| \underline{g^*=\left|p-\frac{i}{2}\right|}}$

Description

Slide 32 of 38 presents the proposed solution to the linear programming problem introduced in the previous slides. The problem involved minimizing $x_1+x_2+x_3+x_4$ subject to constraints derived from approximating a linear function $y = \frac{1}{2}x$ at specific points. Given that $a = \frac{1}{2}$ and $b = 0$, the solution process involves analyzing the double inequalities derived from the absolute value constraints. The analysis leads to $x_1 = x_2 = x_3 = 0$, and $x_4 = |p - \frac{i}{2}|$, where i and p are parameters from the problem statement (not defined on this slide). Therefore, the optimal objective function value, denoted as g^*, is simply $|p - \frac{i}{2}|$. The underlining highlights the "Proposed Solution" and the final result. The context of linear programming is crucial here, as this slide demonstrates the solution methodology for a problem involving absolute value constraints, which requires transformation into a standard linear program before solving.

Figures



(a) Handwritten notes showing the proposed solution to a linear programming problem. The solution involves finding the values of four variables (x_1, x_2, x_3, x_4) that minimize the objective function, given the constraints from previous slides. The solution is derived by analyzing the constraints, which are inequalities involving absolute values. The final solution is given as $x_4 = |p - \frac{i}{2}|$, where i and p are parameters from the problem. The optimal objective function value is $g^* = |p - \frac{i}{2}|$.

Slide 33

Content

$\mathrm{u}_1(t_1 + b \geq 0)$
$u_2(t_1 - b \geq 0)$
$u_3(t_2 + 2a + b \geq 1)$
$u_4(t_2 - 2a - b \geq -1)$
$u_5(t_3 + 4a + b \geq 2)$
$u_6(t_3 - 4a - b \geq -2)$
$u_7(t_4 + ai + b \geq p)$
$u_8(t_4 - ai - b \geq -p) \quad u_i \geq 0$

Description

Slide 33 of 38 presents a system of eight inequalities. Each inequality involves a variable t_i (where $i = 1, 2, 3, 4$) and parameters a, b, i, and p (whose values are defined in previous slides). The inequalities are multiplied by non-negative slack variables u_i, where $i = 1, \ldots, 8$. The inequalities are derived from the absolute value constraints of the linear program presented in previous slides. The context of linear programming suggests that this system of inequalities is a reformulation of the absolute value constraints using slack variables, a common technique in linear programming to convert inequalities involving absolute values into a standard linear programming form. The notation $u_i \geq 0$ indicates that the slack variables are non-negative. The values of a, b, i, and p are not explicitly defined on this slide but are carried over from the previous slides' linear programming problem formulation. This slide likely precedes the dual formulation of the linear program.

Figures

(a) Handwritten notes showing a set of inequalities involving variables t_1, t_2, t_3, t_4 and parameters a, b, i, and p. Each inequality is multiplied by a non-negative variable u_i.

Slide 34

Content

$$
\begin{aligned}
&\max \quad (u_3 - u_4) + 2(u_5 - u_6) + p(u_7 - u_8) \\
&s.t. \quad u_1 + u_2 = 1, \\
&u_3 + u_4 = 1 \\
&u_5 + u_6 = 1 \\
&u_7 + u_8 = 1 \\
&2(u_3 - u_4) + 4(u_5 - u_6) + (u_7 - u_8) = 0 \\
&(u_1 - u_2) + (u_3 - u_4) + (u_5 - u_6) + (u_7 - u_8) = 0 \\
&\underbrace{(u_1 - u_2)}_{V_1} + \underbrace{(u_3 - u_4)}_{V_2} + \underbrace{(u_5 - u_6)}_{V_3} + \underbrace{(u_7 - u_8)}_{V_4} = 0
\end{aligned}
$$

Description

Slide 34 of 38 presents the dual formulation of the linear program from previous slides. The primal problem aimed to approximate a linear function using absolute values. This slide shows the dual problem, which is a maximization problem. The objective function is a linear combination of eight variables ($u_1, \ldots, u_8$), with coefficients derived from the primal problem's constraints. The constraints include four equations where pairs of variables sum to 1 ($u_1 + u_2 = 1$, $u_3 + u_4 = 1$, $u_5 + u_6 = 1$, $u_7 + u_8 = 1$). There's also a linear constraint involving differences between pairs of variables, which is further decomposed into a sum of four terms, each labeled V_i ($i = 1, 2, 3, 4$). The parameter p from the primal problem appears in the objective function. The context of linear programming is crucial here, as this slide demonstrates the dual formulation of a problem, a fundamental concept in linear programming theory and used for solving problems more efficiently or gaining insights into the primal problem's solution.

Figures

$$\max\ (u_3 - u_4)$$

$$\text{s.t.} \quad u_1 + u_2 =$$

(a) Handwritten notes showing a linear programming problem. The objective function is to maximize a linear combination of eight variables $(u_1, u_2, \ldots, u_8)$. The constraints include four equations where pairs of variables sum to 1, and another equation that is a linear combination of the differences between pairs of variables. The last equation is a sum of four terms, each representing the difference between a pair of variables, and each term is labeled V_i for $i = 1, 2, 3, 4$.

Slide 35

Content

$$\begin{cases} u_1(t_1 + b \geq 0) \\ u_2(t_1 - b \geq 0) \\ u_3(t_2 + 2a + b \geq 1) \\ u_4(t_2 - 2a - b \geq -1) \\ u_5(t_3 + 4a + b \geq 2) \\ u_6(t_3 - 4a - b \geq -2) \\ u_7(t_4 + a + b \geq p) \\ u_8(t_4 - a - b \geq -p) \end{cases}$$

a = 1 $\overline{2, \quad b=0 \quad u_i\geq 0}$

Description

Slide 35 of 38 shows a system of eight inequalities, each involving a variable t_i ($i = 1, 2, 3, 4$), parameters a, b, and p, and a non-negative slack variable u_i ($i = 1, \ldots, 8$). The inequalities are multiplied by the corresponding u_i. The values $a = \frac{1}{2}$ and $b = 0$ are given. These inequalities represent a reformulation of the absolute value constraints from the primal linear program presented in previous slides. The red strikethroughs on some terms in the original image suggest that these terms were initially included but later removed or corrected during the problem's formulation. The context of linear programming is crucial, as this slide presents a refined set of constraints likely used in the dual problem formulation or a simplified version of the primal problem after substituting the values of a and b. The non-negativity constraint $u_i \geq 0$ is explicitly stated.

Figures

(a) Handwritten notes showing a set of inequalities involving variables t_1, t_2, t_3, t_4 and parameters a, b, and p. Each inequality is multiplied by a non-negative variable u_i. The values $a = \frac{1}{2}$ and $b = 0$ are specified to the right.

Slide 36

Content

(Cannot say anything about)
($u_1, u_2, u_3, u_4, u_5, u_6$, other than $u_i \geq 0$)
For u_7, u_8 :
$u_7 \left(t_4 + \frac{1}{2} \geq p \right)$
$u_8 \left(t_4 - \frac{1}{2} \geq -p \right)$
If $p \geq \frac{1}{2}$, then $t_4 = p - \frac{1}{2}$, $\implies u_8 = 0, u_7 = 1$
If $p \leq \frac{1}{2}$, then $t_4 = \frac{1}{2} - p \implies u_7 = 0, u_8 = 1$

Description

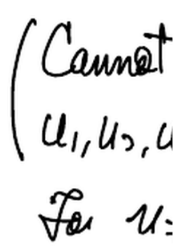
Slide 36 of 38 analyzes the implications of the parameter p on the variables t_4, u_7, and u_8 within the context of the linear program. The slide begins by stating that nothing can be definitively said about the variables u_1 through u_6 except for their non-negativity ($u_i \geq 0$). The focus then shifts to u_7 and u_8, which are associated with inequalities involving t_4 and p. Two cases are considered based on the value of p:

1. **If $p \geq \frac{1}{2}$:** The inequality $t_4 + \frac{1}{2} \geq p$ implies $t_4 \geq p - \frac{1}{2}$. The slide concludes that in this case, $t_4 = p - \frac{1}{2}$, leading to $u_8 = 0$ and $u_7 = 1$.

2. **If $p \leq \frac{1}{2}$:** The inequality $t_4 - \frac{1}{2} \geq -p$ implies $t_4 \geq \frac{1}{2} - p$. The slide concludes that in this case, $t_4 = \frac{1}{2} - p$, leading to $u_7 = 0$ and $u_8 = 1$.

This analysis likely helps to simplify or solve the dual linear program presented in previous slides by determining the optimal values of u_7 and u_8 based on the value of p. The context of linear programming is essential for understanding this analysis of complementary slackness conditions.

Figures



(a) Handwritten notes discussing the implications of the values of p on the values of t_4, u_7, and u_8.

Slide 37

Content

For $\text{p} \geq \frac{1}{2}$,

$\max \quad V_2 + 2V_3 + p$

$s.t. \quad 2V_2 + 4V_3 + 1 = 0$

$V_1 + V_2 + V_3 + 1 = 0$

$\max \quad p - \frac{1}{2}$

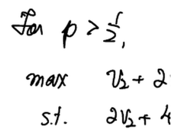
$s.t. \quad V_2 + 2V_3 = -\frac{1}{2}$

$V_1 + V_2 + V_3 = -1$

Description

Slide 37 of 38 presents two linear programming problems. The first problem is a maximization problem where the objective function is $V_2 + 2V_3 + p$, subject to two linear equality constraints: $2V_2 + 4V_3 + 1 = 0$ and $V_1 + V_2 + V_3 + 1 = 0$. The second problem, indicated by a red arrow, is also a maximization problem, but the objective function is $p - \frac{1}{2}$, subject to two different linear equality constraints: $V_2 + 2V_3 = -\frac{1}{2}$ and $V_1 + V_2 + V_3 = -1$. A handwritten note next to the second problem emphasizes the importance of checking its feasibility. The condition "For $p \geq \frac{1}{2}$" suggests that these formulations are valid only when p is greater than or equal to $\frac{1}{2}$. The context of linear programming is crucial here, as the slide shows two different formulations of the same problem (or related problems) likely derived from the dual problem or a simplification of the primal problem, aiming to find the optimal solution under specific conditions on p. The variables V_i are likely derived from the differences of the u_i variables from previous slides.

Figures



(a) Handwritten notes showing two linear programming problems. The first problem maximizes a linear combination of V_2, V_3, and p, subject to two linear constraints. The second problem maximizes $p - \frac{1}{2}$, subject to two other linear constraints. A red arrow points to the second problem, and a note "Make sure this is feasible" is written next to it.

Slide 38

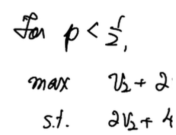
Content

For $p \leq \frac{1}{2}$,
max $\quad V_2 + 2V_3 - p$
$s.t. \quad 2V_2 + 4V_3 - 1 = 0$
$V_1 + V_2 + V_3 - 1 = 0$
max $\quad \frac{1}{2} - p$
$s.t. \quad V_2 + 2V_3 = \frac{1}{2}$
$V_1 + V_2 + V_3 = 1$

Description

Slide 38 of 38 presents two linear programming problems, similar to slide 37 but for the case when $p \leq \frac{1}{2}$. The first problem maximizes $V_2 + 2V_3 - p$ subject to the constraints $2V_2 + 4V_3 - 1 = 0$ and $V_1 + V_2 + V_3 - 1 = 0$. The second problem, indicated by a red arrow, maximizes $\frac{1}{2} - p$ subject to $V_2 + 2V_3 = \frac{1}{2}$ and $V_1 + V_2 + V_3 = 1$. A handwritten note emphasizes checking the feasibility of the second problem. These formulations are likely derived from the dual problem or a simplification of the primal problem, aiming to find the optimal solution under the condition $p \leq \frac{1}{2}$. The variables V_i are likely derived from the differences of the u_i variables from previous slides. The condition "For $p \leq \frac{1}{2}$" indicates that these formulations are valid only when p is less than or equal to $\frac{1}{2}$.

Figures



(a) Handwritten notes showing two linear programming problems. The first problem maximizes a linear combination of V_2, V_3, and p, subject to two linear constraints. The second problem maximizes $\frac{1}{2} - p$, subject to two other linear constraints. A red arrow points to the second problem, and a note "Make sure this is feasible" is written next to it.